A Parameter and Flag Adaptive Reconstruction Method for Satellite Vegetation Index Time Series

Huanfeng Shen, *Senior Member, IEEE*, Yuxi Ran, Xiaobin Guan, *Member, IEEE*, Dong Chu, and Dongyi Li

***Abstract*— The data quality issue induced by atmospheric and other disturbances can significantly impede the application of remote sensing vegetation indices (VIs). Despite the development of numerous VI reconstruction techniques, two major challenges remain, that is, the reliance on quality flag inputs and parameter settings. Quality flags are usually necessary as inputs for the different methods to improve the reconstruction accuracy, but mislabeling can be common in the quality flag data, which can directly introduce uncertainties. Furthermore, constant parameter schemes are usually assigned during the reconstruction applications, but the optimal parameters for all the models can show great spatial heterogeneity. Accordingly, in this article, we propose a parameter-free and flag-free adaptive time-series method based on a variational reconstruction framework (PF-Free) to address the aforementioned issues, which can be applied without any parameter or flag inputs. PF-Free makes full use of the time-series temporal smoothness and interannual similarity to label the data after time-series rearrangement, and the parameters are adaptively selected using an improved generalized cross-validation (GCV) technique. Simulation and real-data experiments demonstrate that PF-Free can achieve better and more stable reconstruction results, compared to other comparative methods. The adaptive quality flags can denote the data quality robustly and accurately, while guaranteeing better reconstruction performance, as long as there is no mislabeling in the original flags. Moreover, the adaptive parameter selection strategy considers the great spatial heterogeneity in the optimal parameters on a pixel-by-pixel basis, leading to more stable reconstruction outcomes under complex conditions. Further experiments also prove the effectiveness of PF-Free in processing data without quality flags or severely contaminated data, using Advanced Very High-Resolution Radiometer (AVHRR) data and Moderate Resolution Imaging Spectroradiometer (MODIS) daily normalized difference vegetation index (NDVI) data. This work provides a practical reconstruction method for the VI time series, which is both flexible and convenient, without requiring any parameter or flag input, which we believe will advance VI reconstruction applications.**

***Index Terms*— Adaptive, generalized cross-validation (GCV), quality flags, temporal reconstruction, variational regularization, vegetation indices (VIs).**

I. Introduction

REMOTE sensing vegetation indices (VIs) are a means of mathematically combining two or more spectral bands to reveal the status and growth of vegetation [1]. Taking the most widely used normalized difference vegetation index (NDVI) as an example [2], which is obtained from the simple calculation of red and near-infrared reflectance, this can be used in the fields of vegetation dynamics monitoring, productivity estimation, and so on [3], [4]. However, the VI time series is often disturbed by clouds, aerosols, and other adverse atmospheric conditions, resulting in unreasonably sudden drops in their values or even missing data [5]. Although maximum-value composite (MVC) preprocessing is usually applied in most of the VI products [6], there can still be residual noise and gaps in the time series, which significantly limit their further application [7], [8]. Therefore, it is necessary to reconstruct VIs to obtain high-quality and gap-free VI time-series data [9], [10].

A series of VI reconstruction methods have already been developed, which can be broadly categorized into two main groups: temporal domain-based methods and combined spatiotemporal domain-based methods [11]. The combined spatiotemporal domain-based methods, such as the spatial–temporal Savitzky–Golay (STSG) method [12] and the spatiotemporal tensor completion (ST-Tensor) method [13], integrate the temporal continuity and spatial correlation of VI time series and can achieve superior performance in scenarios with continuous missing data. However, such methods typically entail a high computational cost and high hardware requirements, which limits their practical application when processing huge datasets. Therefore, the efficient temporal-domain-based methods still remain the mainstream in VI reconstruction applications, as they address the temporal continuity of the VI time series on a pixel-by-pixel basis [14], and

Received 27 December 2024; revised 8 April 2025; accepted 4 May 2025. Date of publication 9 May 2025; date of current version 29 May 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42371364 and Grant U23A2021 and in part by China Postdoctoral Science Foundation Special Project under Grant 2022T150489. *(Corresponding author: Xiaobin Guan.)*

Huanfeng Shen and Xiaobin Guan are with the School of Resource and Environmental Sciences, Wuhan University, Wuhan 430079, China, also with the Key Laboratory of Geographic Information System, Ministry of Education, Wuhan 430079, China, and also with the Key Laboratory of Digital Mapping and Land Information Application, Ministry of Natural Resources, Wuhan 430079, China (e-mail: shenhf@whu.edu.cn; guanxb@whu.edu.cn).

Yuxi Ran and Dongyi Li are with the School of Resource and Environmental Sciences, Wuhan University, Wuhan 430079, China (e-mail: yx.ran@whu.edu.cn; DongyiLi@whu.edu.cn).

Dong Chu is with the Key Laboratory of Earth Surface Processes and Regional Response in the Yangtze-Huaihe River Basin, School of Geography and Tourism, Anhui Normal University, Wuhu, Anhui 241002, China (e-mail: chudong@ahnu.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2025.3568444, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2025.3568444

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

thus can achieve improved efficiency. The temporal methods can be classified into four main categories [15], that is, function fitting methods [16], [17], local window filtering methods [18], [19], frequency-domain methods [20], [21], and other methods such as the methods based on the principle of variational regularization [22], [23]. Each of these four categories has its own strengths and limitations, making them suitable for different types of applications. Although these temporal-domain-based methods have achieved relatively satisfactory reconstruction results, there are still two main issues that can limit their performance in practical applications, that is, the data quality flags and parameter settings.

First of all, most of the temporal methods necessitate the utilization of data quality flags to achieve enhanced reconstruction outcomes [23]. The quality flags can provide information reflecting the reliability of the data for reconstruction, allowing reconstruction using only the trusted information and avoiding the introduction of additional errors. Although some temporal methods, such as the best index slope extraction (BISE) method [24], can be conducted without quality flags, they are prone to problems such as underfitting and incorrect reconstruction, and the use of flags will result in higher accuracy [25]. However, it has been shown that inappropriate quality flags can have a significant impact on the reconstruction results [26], [27], especially in scenarios where the gaps are labeled as valid data (i.e., omission of cloud flags), injecting erroneous references into the reconstruction [28], [29]. Many studies have already indicated that there are uncertainties in the flags provided by the widely used Moderate Resolution Imaging Spectroradiometer (MODIS) data [30], which would certainly affect the quality of the results if these data were used directly for reconstruction [31], [32]. Dense VI time series, such as daily data or even higher frequency data, typically contain substantial noise and gaps, so the reliability of the flags becomes even worse. Moreover, some products do not even come with flags, such as the Global Inventory Modeling and Mapping Studies-Third Generation V1.2 data for NDVI from the Advanced Very High-Resolution Radiometer (AVHRR GIMMS NDVI3g+), which can only be reconstructed without flags [33]. If data quality flags are necessary for optimal reconstruction results, these unlabeled data become difficult to apply. Therefore, there is an urgent need to develop methods that do not require the input of quality flags, to boost the VI reconstruction performance in different situations.

Furthermore, reasonable and accurate parameter settings are crucial to the ability of the reconstruction methods [34], [35], especially in large-scale applications when the parameter selection has a considerable influence on the results. There are key parameters controlling the reconstruction results for almost all the VI reconstruction methods. For instance, the local window filtering methods regulate the smoothness of the reconstructed curves by varying the radius of the sliding window [36], while the frequency-domain methods require the setting of parameters such as the frequency magnitude and fitting threshold to achieve a balance between robustness and fidelity [27]. In addition, the regularization parameter λ of the variational methods is employed to control the degree of smoothness of the reconstructed sequence. A larger (or smaller) λ results in a smoother (or coarser) reconstruction [37], [38]. The settings of the above parameters should vary in different regions due to the growth dynamics of vegetation and the degree of noise and gaps [22]; however, constant parameters are usually assigned to the different methods in various applications [39]. For example, in regions with different vegetation types, the VI time series can be different. In this condition, if the same constant parameters are used, large uncertainties will consequently be induced in the reconstruction results. At a regional or global scale, the variations in the reconstruction parameters of VI sequences will be even more complex, since differences in parameters can even be found in neighboring pixels [40]. As a result, although the assignment of constant parameters is convenient, it will inevitably lead to poor application effectiveness in certain areas.

Hence, it is necessary to address the above two issues regarding quality flags and parameter settings, to promote VI time-series reconstruction accuracy in large-scale applications. In most of the current reconstruction applications, a flag time series is also required as input, together with the VI time series. As a result, the additional data input can induce errors in the reconstruction results. Therefore, if we can obtain high-accuracy quality flags based on the VI time series itself, it will become possible to further advance the reconstruction accuracy without any additional flag inputs, which will greatly facilitate the VI reconstruction applications [5], [41]. In the case of reconstruction parameters, the most satisfactory scenario is that we do not need to preset any parameter for the reconstruction method, and we can adaptively obtain the optimal parameters according to the VI time series itself [42], [43], which would greatly improve the simplicity and convenience [44], [45].

This study aimed to develop an adaptive reconstruction method for VI time series that is both parameter-free and flag-free (PF-Free), as a possible solution to advance the reconstruction of VI time series at large scales. PF-Free integrates the temporal smoothness and interannual similarity of the VI time series to generate reliable quality flags through rearrangement, based on which reconstruction parameters are selected at the pixel level using an improved generalized cross-validation (GCV) technique. In the following, we provide a detailed description of the implementation principle of PF-Free. We then describe the evaluation of the performance of the proposed method through comparison experiments with other reconstruction methods and finally discuss its advantages and limitations.

II. Study Areas and Data

In this study, six representative MODIS tile regions (h12v04, h13v09, h18v03, h19v08, h27v04, and h27v06) with different noise and gap conditions were selected as the study areas. The specific information and locations of these regions are presented in Fig. 1 and Table I in the supplementary material [46], [47]. Most of the areas in the six tiles are frequently contaminated by clouds [48], and the vegetation types are diverse, making these tiles well-suited for experimentation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

TABLE I
RULES FOR QUALITY FLAG GENERATION

Spatial relationship between the data point and confidence intervals	Quality status of the data point	Quality flag
Inside the inner confidence interval	Good	0
Outside the inner confidence interval and inside the outer confidence interval	Uncertain	1
Outside the outer confidence interval	Bad	3

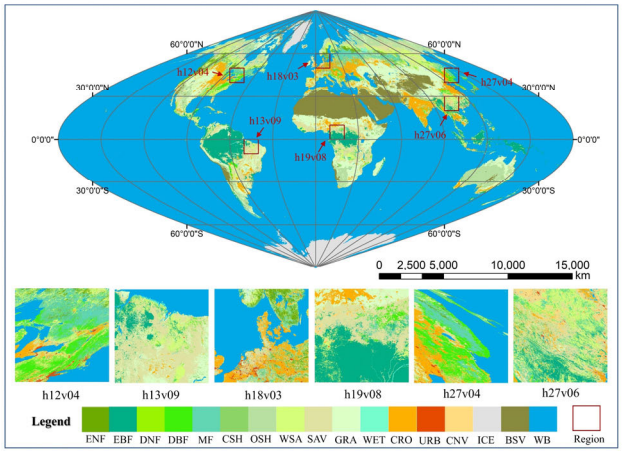


Fig. 1. Global distribution of the six representative MODIS tile regions. ENF, evergreen needleleaf forests; EBF, evergreen broadleaf forests; DNF, deciduous needleleaf forests; DBF, deciduous broadleaf forests; MF, mixed forests; CSH, closed shrublands; OSH, open shrublands; WSA, woody savannas; SAV, savannas; GRA, grasslands; WET, permanent wetlands; CRO, croplands; URB, urban and built-up lands; CNV, cropland/natural vegetation mosaics; ICE, permanent snow and ice; BSV, barren; WB, water bodies.

The main data source used in this study was MODIS data, with 16-day NDVI data as the main experimental data, and daily NDVI data used to test the performance of PF-Free in scenarios where poor data quality and labeling difficulties exist. The 16-day NDVI was the MODIS version 6 NDVI dataset (MOD13A2) for 2003–2020 [49] with a 1-km resolution, which is freely available from the national aeronautics and space administration (NASA) website (https://search.earthdata.nasa.gov/search). This dataset is accompanied by quality assurance (QA) information that classifies the NDVI into five categories with the help of a pixel reliability index (RI): good (RI = 0), uncertain (RI = 1), ice/snow (RI = 2), cloudy (RI = 3), and filled/no data (RI = −1) [5], [50]. The daily NDVI was derived from the MODIS daily surface reflectance dataset (MOD09GA) by band calculation [51] (hereafter referred to as MOD09GA-NDVI). Due to the lack of MVC preprocessing in the MOD09GA dataset, it contains a large amount of noise and gaps, resulting in the poor quality of the MOD09GA-NDVI. The data above have been processed with corrections for the atmosphere and geometry. Vegetation-type data were also used in this study, which was derived from the 500-m resolution MCD12Q1 vegetation classification dataset and employed the international geosphere-biosphere programme (IGBP) classification scheme.

Moreover, AVHRR GIMMS NDVI 3g+ data (free on https://doi.org/10.3334/ORNLDAAC/2187) were also included [52], [53]. These data feature a temporal resolution of 15 days and a spatial resolution of 1/12°. Nevertheless, they lack data quality flags [33], which poses a challenge to their reconstruction.

III. METHODOLOGY

A. Framework of PF-Free

In this article, the raw VI time series is denoted as $y = [y_1, y_2, \ldots, y_t, \ldots, y_n]^{\mathrm{T}} \in \boldsymbol{R}^{n\times 1}$, while the cleanly reconstructed VI time series is denoted as $x = [x_1, x_2, \ldots, x_t, \ldots, x_n]^{\mathrm{T}} \in \boldsymbol{R}^{n\times 1}$, where n is the length of the time series. The objective of VI time-series reconstruction is to obtain the desired x from the original y. We referred to the framework of [1] and revised the regularization terms to conduct reconstruction using the following variational model:

$$\hat{x} = \underset{x}{\operatorname{argmin}}\, W\|x-y\|_2^2 + \lambda_1\|Dx\|_2^2 + \lambda_2\|Lx\|_2^2 \tag{1}$$

where $W \in \boldsymbol{R}^{n\times n}$ denotes the diagonal matrix, and the main diagonal elements of W are the weights of each VI point determined from the quality flags. $D \in \boldsymbol{R}^{(n-2)\times n}$ and $L \in \boldsymbol{R}^{(n-T)\times n}$ represent regularization matrices named the second-order difference matrix and interannual difference matrix, respectively, which are defined as shown in the following equations:

$$D = \begin{bmatrix} 1 & -2 & 1 & & & & & \\ 0 & 1 & -2 & \cdots & & 0 & 0 & 0 \\ & \vdots & & \ddots & & \vdots & & \\ 0 & 0 & 0 & & -2 & 1 & 0 & \\ 0 & 0 & 0 & \cdots & 1 & -2 & 1 & \end{bmatrix} \tag{2}$$

$$L = \begin{bmatrix} 1 & 0 & \cdots & 0 & -1 & 0 & \ldots & 0 \\ 0 & 1 & \cdots & 0 & 0 & -1 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 1 & 0 & 0 & \ldots & -1 \end{bmatrix}. \tag{3}$$

Here, λ_1 and λ_2 are regularization parameters. Equation (1) contains three terms, namely, the fidelity term, the temporal smoothness term, and the interannual similarity term. The fidelity term ensures that the reconstructed series does not deviate excessively from the original series. The temporal smoothness term characterizes the local gradual change in vegetation growth, which is manifested in the tendency of the second-order difference of the time series $x_{t-1} - 2x_t + x_{t+1}$ to tend to zero [10]. The interannual similarity term embodies the cyclical feature of phenology-driven vegetation growth. Specifically speaking, the values of the same time period in neighboring years tend to be close to each other, that is,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

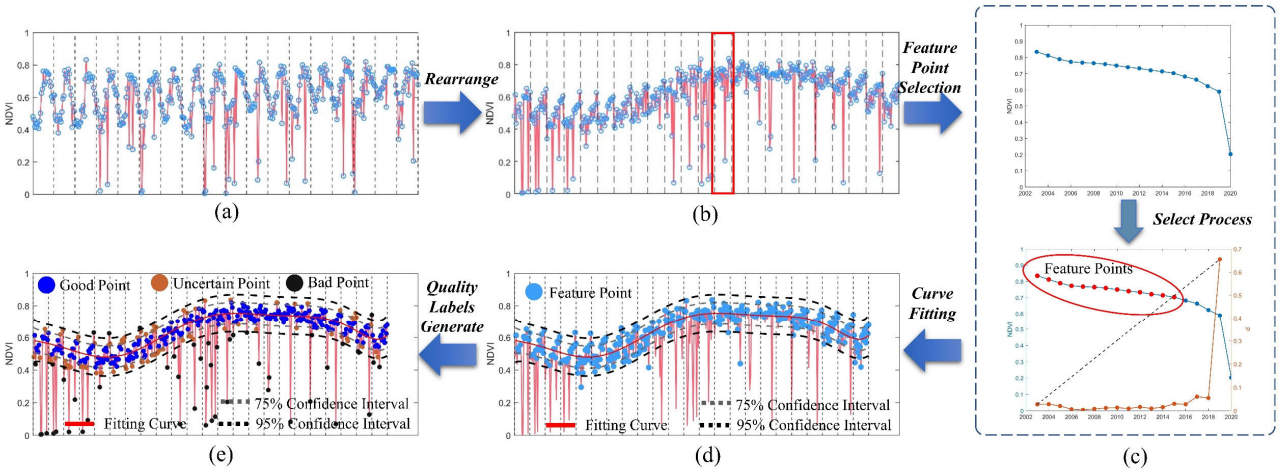


Fig. 2. Flowchart of adaptive flag generation. (a) Raw VI (NDVI) time series. (b) Rearranged VI time series. (c) Example of feature point screening for the data in the red box in (b). (d) Feature points, fitting curves, and confidence intervals of the rearranged NDVI time series. (e) Quality flag generation by determining the spatial relationship between the data points and confidence intervals.

$x_t - x_{t+T}$ converges to zero, assuming T is the length of the one-year VI time series [54].

The PF-Free method reconstructs the VI time series through the following three steps: 1) input the raw VI time series and adaptively generate the quality flags; 2) select the optimal regularization parameters by the improved GCV strategy; and 3) solve (1) based on the first two steps to calculate the reconstructed VI time series. In the following, the first two steps are described in detail.

B. Adaptive Flag Generation

To complete the reconstruction in scenarios where quality flags are missing or imprecise, an adaptive flag generation method for the VI time series is proposed, which requires only the input of the VI time series and directly generates the corresponding quality flags. In principle, this method assumes that the data in the same time period over many years are usually similar and that values far away from the multiyear average are of relatively lower quality. Based on the above assumptions, the method automatically generates quality flags through three steps of data rearrangement, feature point screening, and curve fitting, which are described in depth using a real VI time series as an example in Fig. 2.

1) Data Rearrangement: The raw VI time series y is not an optimal representation of the distribution characteristics of the multiyear data from the same time period [see Fig. 2(a)] and thus it is necessary to rearrange the data. Specifically, y is first converted into a 2-D matrix Y, where each row of Y consists of the same-period data points in each year. Then, Y is dimensionally reduced to obtain the rearranged time series y^*. This process is illustrated in Fig. 3. When y^* is plotted [see Fig. 2(b)], it is clearly evident that y^* reflects the average trend of the vegetation phenology corresponding to the VI time series in one year. If a polynomial can be fit to the data points on y^*, and the fit curve can be used to characterize the vegetation phenology, then the quality of the data points can be assessed by the spatial relationship between the data points on y^* and the fit curve.

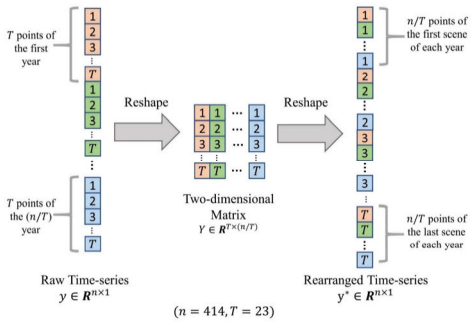


Fig. 3. Flowchart of data rearrangement for vegetation phenology pattern reflection.

2) Feature Point Screening: Not all points can be used to fit the curve as gap points introduce errors. Therefore, it is necessary to screen for points with no or few gaps (hereafter referred to as feature points). To illustrate this, consider the red-boxed time phase in Fig. 2(b). Initially, all the points are sorted in descending order, and then the decline magnitude values d_t for all the points are computed by the following equation:

$$d_t = \left(y_t^* - y_{t+1}^*\right)/y_t^* \tag{4}$$

where y_t^* and y_{t+1}^*, respectively, denote the tth and $(t+1)$th rearranged data (descending order, $y_t^* > y_{t+1}^*$), while d_t denotes the values of the decline magnitude. The declining curve is obtained by connecting the d_t of all the points [brown curve in Fig. 2(c)]. Noise and gaps in the VI time series tend to be negatively biased, leading to larger d_t values. Consequently, with the initial point and highest d_t point joined [the dotted line in Fig. 2(c)], we calculate the distance from each point to this dotted line. For robustness, n points with the largest distance are selected ($n = 4$ was determined experimentally). The point with the highest value sets the lower boundary, and this point, along with the higher values, is selected as the feature point [the red dots in Fig. 2(c)]. Fig. 2(d) illustrates the results of the feature point screening (blue points) of a VI time series, showing a significant reduction in the number of deletions among these points.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

TABLE II
WEIGHTING SCHEMES FOR THE DIFFERENT METHODS

Method	Quality flag	Weight
IDR, SG, HANTS	0, 1	1
	−1, 2, 3	0
WS, Season-L1, PF-Free	0	1
	1	0.8
	−1, 2, 3	0

3) Curve Fitting: Next, the curve is fit utilizing the feature points, with the order of the fit curve being dynamically adjusted according to the proportion of feature points in the time series, as shown in Table II in the supplementary material. Thereafter, we employ the fit curve to create a 75% inner confidence interval and a 95% outer confidence interval. The resulting intervals are illustrated in Fig. 2(c). We believe that the above intervals reflect the normal variation of VI values. Points falling within the intervals are considered to be of superior quality, whereas those outside the intervals are considered to be of inferior quality. The points are labeled according to the specific rules given in Table I. The flags generated in this process are shown in Fig. 2(e). Using this methodology, the majority of the points in the VI time series can be accurately labeled.

C. Adaptive Parameter Selection

In this study, an improved GCV strategy was employed for the adaptive selection of regularization parameters. GCV determines the optimal regularization parameters by minimizing the GCV scores [55], obviating the need for knowledge of the mean and variance of the data. GCV has been widely used in the fields of image restoration, super-resolution reconstruction, and so on [2]. Thus, it can also be applied to the field of VI time-series reconstruction to achieve adaptive parameter selection. GCV is solved by the following equation [3]:

$$\begin{aligned}\lambda_i &= \operatorname{argmin}\left(\mathrm{GCV}\left(\lambda_i\right)\right)\\ &= n\left\|\left(I - H\left(\lambda_i\right)\right)y\right\|^2 / \operatorname{trace}\left(I - H\left(\lambda_i\right)\right)^2\end{aligned} \tag{5}$$

where $i = 1, 2$, and $H(\lambda_i)$ is the projection matrix of the original NDVI time series projected into (1), which is calculated as shown in (6). trace() is the trace of the calculated matrix, regularization matrix C is the second-order difference matrix D when $i = 1$, and the interannual difference matrix L when $i = 2$

$$H\left(\lambda_i\right) = \left(W + \lambda_i C^{\mathrm{T}} C\right)^{-1}. \tag{6}$$

This improved GCV strategy was used to select the regularization parameters. This is because empirical evidence has demonstrated that the original GCV method is prone to instability when dealing with multiple parameter determination, rendering it challenging to obtain the optimal parameters simultaneously [4]. To circumvent this issue, we enhanced the stability of the GCV parameter estimation by reducing the number of parameters. The following relationship is postulated between the two regularization parameters:

$$\lambda_2 = k\lambda_1 = k\lambda \tag{7}$$

$$k = \|Dx\|_2^2 / \|Lx\|_2^2 \tag{8}$$

where λ represents λ_1, and k is the scaling factor between the two regularization parameters. This factor reflects the strength of the relationship between the ideal sequence in the two regularization terms of temporal smoothness and interannual similarity. Consequently, it can also be used to measure the relationship between the two regularization parameters.

In this study, the reconstructed time series in (8) was initially obtained through the Whittaker smoother (WS) [22]. Similar to the PF-Free method, WS is based on the principle of variational regularization and employs the temporal smoothness constraint to conduct denoising and gap-filling, demonstrating remarkable performance in both reconstruction and efficiency. It was then treated as an ideal sequence, and the scale factor k was determined by (8). It is evident that the process of determining the scale factor k is inherently adaptive, allowing for the identification of a unique k value for each pixel. The refined model (9) is derived by integrating (1) and (8). The model requires the estimation of the scale factor k and the regularization parameter λ, which mitigates the uncertainty in GCV as it adapts to the parameters

$$\hat{x} = \underset{x}{\operatorname{argmin}}\, W\|x - y\|_2^2 + \lambda\|Dx\|_2^2 + k\lambda\|Lx\|_2^2. \tag{9}$$

IV. EXPERIMENTS AND RESULTS

The PF-Free method was compared with five commonly used temporal domain-based methods: the iterative interpolation for data reconstruction (IDR) method [19], the Savitzky–Golay (SG) filter [18], the harmonic analysis of time-series method (HANTS) [59], the WS [22], and the Season-L1 method [1]. These five comparison methods have been widely used in related studies. The parameter settings of the comparison methods are provided in Tables III–VII in the supplementary material. The weighting schemes for the above six methods are shown in Table II, where the quality flags for the five comparison methods were derived from the RI data provided by the MODIS products.

A. Simulation Experiments

Simulation experiments based on the 16-day MODIS NDVI data were conducted to quantitatively evaluate the performance of the proposed PF-Free method. Since completely clean NDVI time series do not exist in reality, the reference NDVI time series was generated and treated as real data reflecting the vegetation growth pattern [13], [40]. Briefly, a one-year reference series was generated by first calculating the average of the valid data for the same period for all the years, and then variations were applied to it to obtain a reference series reflecting different interannual trends. The simulated NDVI series was generated by combining the reference NDVI series with the real quality flags. Specifically, when RI = 1, the corresponding NDVI data points were considered as noise points with values ranging from 90% to 105% of the reference NDVI value; when RI = 2 or 3, the data points were considered as gap points with values not exceeding 10% of the reference NDVI value. In addition, taking into account the existence of mislabeling in the real quality flags, we simulated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

TABLE III
SIMULATION SCHEME FOR THE REFERENCE NDVI TIME SERIES

Simulation points	Simulation approach	Note
Noise points	Ranging from 90% to 105% of the reference NDVI value	RI needs to be 1
Gap points	Not exceeding 10% of the reference NDVI value	RI needs to be 2 or 3
Mislabeled points	Change the RI of 10% of the gap points (RI = 3) to 0	Only points in the specified 1/4 sub-region are mislabeled

the erroneous flag circumstances accordingly. That is, a 1/4 specified subregion was selected on each tile region, and 10% of the gap points (RI = 3) caused by clouds were randomly selected within the subregion and manually labeled as valid points (RI = 0), while those outside the subregions were not modified. These simulation rules are summarized in Table III. After the reconstruction was completed, the mean absolute error (MAE) and correlation coefficient (CC) by (10) and (11) were calculated to evaluate the accuracy and curve trend of these methods

$$\mathrm{MAE} = \sum_{i=1}^{n} \left| x_i - \hat{x_i} \right| / n \tag{10}$$

$$\mathrm{CC} = \frac{\sum_{i=1}^{n} (x_i - \mu_x)\left(\hat{x}_i - \mu_{\hat{x}}\right)}{\sqrt{\sum_{i=1}^{n} (x_i - \mu_x)^2} \sqrt{\sum_{i=1}^{n} \left(\hat{x}_i - \mu_{\hat{x}}\right)^2}}. \tag{11}$$

The reconstruction results of the different methods on each tile region are shown in Tables IV and V. Among the different methods, IDR performs the worst, with the highest MAE and the lowest CC. This is followed by the HANTS method and the SG filter. The SG filter can quickly approach the upper envelope of the NDVI series during the reconstruction process, and thus has a strong noise reduction capability, while the HANTS method has a more robust gap-filling ability. The variational WS filtering method, the Season-L1 method, and the PF-Free method achieve even better results in the simulation experiments. However, the comparison methods are affected by mislabeling, and the MAE and CC values are almost all worse than those of PF-Free. This can be attributed to PF-Free obtaining the best solution due to its automatic generation of quality flags and parameters. Accurate quality flags can correctly guide the reconstruction process, and the adaptive reconstruction parameters increase the robustness of a wide range of reconstruction tasks.

Figs. 4 and 5 show the spatial distribution of the MAE on the reconstructed noise points and gap points for the different methods, where the region inside the closed box is the mislabeled subregion and outside of this is the correctly labeled region. It can be seen that, in the correctly labeled region, the accuracy of the IDR method is the worst, which is consistent with the conclusions reflected in Tables IV and V. The five remaining methods show little difference in the noise points, and the gap point situation indicates that the variational methods which are based on the principle of variational regularization (WS filtering method, Season-L1 method, and PF-Free method) perform better than the nonvariational methods (IDR method, HANTS method, and SG filter). However, in the mislabeled subregion, the accuracy of the comparison methods shows a significant decrease, with the gap points decreasing more than the noise points, and the variational methods decreasing more than the nonvariational methods. In particular, the WS method is the most affected due to the different ways of using flags between the variational and nonvariational methods, where the former needs to incorporate flags in the variational solving process, and the latter only needs to use them to filter data during the linear interpolation process. The proposed PF-Free method, which is also based on the variational principle, obtains the lowest MAE because it generates quality labels of high confidence with the help of the self-labeling step, thus avoiding misleading reconstruction by false quality flags.

B. Real Data Experiments

1) Reconstruction Evaluation of Temporal Curves: To verify the performance of the proposed PF-Free method in reconstructing real NDVI time series, eight vegetation pixels (covering the main vegetation types in the study regions) were selected and reconstructed using the PF-Free method as well as the SG filter, HANTS method, and Season-L1 method. Fig. 6 shows the original NDVI curves for the above eight pixels and the reconstruction results of the different methods. In general, PF-Free shows a better performance in noise smoothing and gap filling, which can be summarized as the following three main advantages. First, the proposed PF-Free method can deal with scenarios with consecutive gap cases [e.g., Fig. 6(a)–(g)]. Taking Fig. 6(e) as an example, there is a long continuous gap in 2012, which is located at the peak of the time series. Both the SG filter and the HANTS method are unable to reconstruct the trajectory of the NDVI in the waveform peak, and only the PF-Free method and the Season-L1 method successfully recover the wave crest curve using the valid information from adjacent years. Next, the PF-Free method can avoid being misled by inappropriate quality flags. The situation where low-quality points are labeled as high-quality points is shown in Fig. 6(c), (d), (f), and (h), which results in all three comparison methods using original flags being misguided by the unreasonable flags, especially the SG filter, which results in the reconstruction of the most residual noise and clearly deviates from the trajectory of normal growth of NDVI. In contrast, PF-Free generates the corresponding

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
MAE AND CC OF THE VARIOUS METHODS FOR NOISE POINTS ON EACH TILE REGION IN THE SIMULATION EXPERIMENTS

		IDR	HANTS	SG	WS	Season-L1	PF-Free
MAE	h12v04	0.0228	0.0173	0.0127	0.0136	0.0133	**0.0069**
	h13v09	0.0270	0.0168	0.0140	0.0199	0.0158	**0.0112**
	h18v03	0.0234	0.0132	0.0115	0.0148	0.0134	**0.0102**
	h19v08	0.0261	0.0168	**0.0137**	0.0210	0.0164	0.0139
	h27v04	0.0195	0.0160	0.0107	0.0116	0.0120	**0.0103**
	h27v06	0.0258	0.0173	**0.0143**	0.0188	0.0163	0.0155
	mean	0.0241	0.0162	0.0128	0.0166	0.0145	**0.0113**
CC	h12v04	0.9714	0.9805	0.9882	0.9814	0.9874	**0.9930**
	h13v09	0.8580	0.9269	0.9233	0.8606	0.9197	**0.9578**
	h18v03	0.9190	0.9682	0.9695	0.9423	0.9658	**0.9832**
	h19v08	0.8947	0.9350	0.9339	0.8448	0.9350	**0.9587**
	h27v04	0.9888	0.9929	**0.9957**	0.9934	0.9956	0.9940
	h27v06	0.9339	0.9662	0.9649	0.9205	0.9687	**0.9778**
	mean	0.9276	0.9616	0.9626	0.9238	0.9620	**0.9774**

TABLE V
MAE AND CC OF THE VARIOUS METHODS FOR GAP POINTS ON EACH TILE REGION IN THE SIMULATION EXPERIMENTS

		IDR	HANTS	SG	WS	Season-L1	PF-Free
MAE	h12v04	0.0863	0.0790	0.0865	0.0392	0.0243	**0.0213**
	h13v09	0.0278	0.0200	0.0282	0.0285	0.0265	**0.0121**
	h18v03	0.0383	0.0217	0.0375	0.0239	0.0205	**0.0127**
	h19v08	0.0371	0.0330	0.0365	0.0418	0.0398	**0.0137**
	h27v04	0.0608	0.0602	0.0602	0.0272	0.0174	**0.0151**
	h27v06	0.0314	0.0264	0.0319	0.0331	0.0310	**0.0148**
	mean	0.0470	0.0401	0.0468	0.0323	0.0266	**0.0150**
CC	h12v04	0.4306	0.5691	0.4422	0.8057	0.9005	**0.9444**
	h13v09	0.7010	0.8124	0.7039	0.7772	0.7307	**0.9115**
	h18v03	0.5169	0.8305	0.5476	0.7833	0.8147	**0.9409**
	h19v08	0.7257	0.7347	0.7269	0.7316	0.6806	**0.9047**
	h27v04	0.7426	0.8153	0.7562	0.9153	0.9606	**0.9817**
	h27v06	0.7938	0.8455	0.7781	0.8045	0.8069	**0.9441**
	mean	0.6518	0.7679	0.6592	0.8029	0.8157	**0.9379**

markers directly from the original NDVI curves, avoiding the effect of using incorrect quality flags. Finally, PF-Free can better achieve a balance between noise removal and feature preservation, as shown in Fig. 6(b) and (c). The results of PF-Free do not contain residual noise, while also retaining the normal fluctuation of the temporal curves, highlighting the peaks and valleys of vegetation growth and other key points. This can be attributed to the fact that the PF-Free method realizes the automatic selection of reconstruction parameters through the improved GCV strategy. The process of selection is at the pixel level, that is, each NDVI curve corresponds to the optimal reconstruction parameters. Therefore, PF-Free can perform the task of curve reconstruction for different noise levels and vegetation types.

2) Reconstruction Evaluation of Spatial Images: Fig. 7 compares the spatial distribution of the original NDVI and the NDVI reconstructed by the different methods in the h27v06 tile region for DOY 1 in 2012. Affected by cloud contamination, there is a large range of low-value pixels in the original NDVI image, while the reconstruction results of the different methods are similar when considered as a whole.

However, there are differences in the two localized regions of A and B. The enlarged images of regions A and B in Fig. 7 are shown in Fig. 8.

Region A is mainly located in the western part of Chongqing, China, where dense cloud cover reduces the NDVI values in this region, as can be seen in Fig. 8(a). The results of the SG filter and HANTS method suffer from overestimation, and the smoothing is incomplete in some areas [see the black circles in Fig. 8(a)]. In comparison, the Season-L1 method and PF-Free method are relatively consistent, but there are some anomalous fluctuations in the results of the Season-L1 method, which are not spatially continuous enough. Only PF-Free performs better in terms of noise reduction and spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

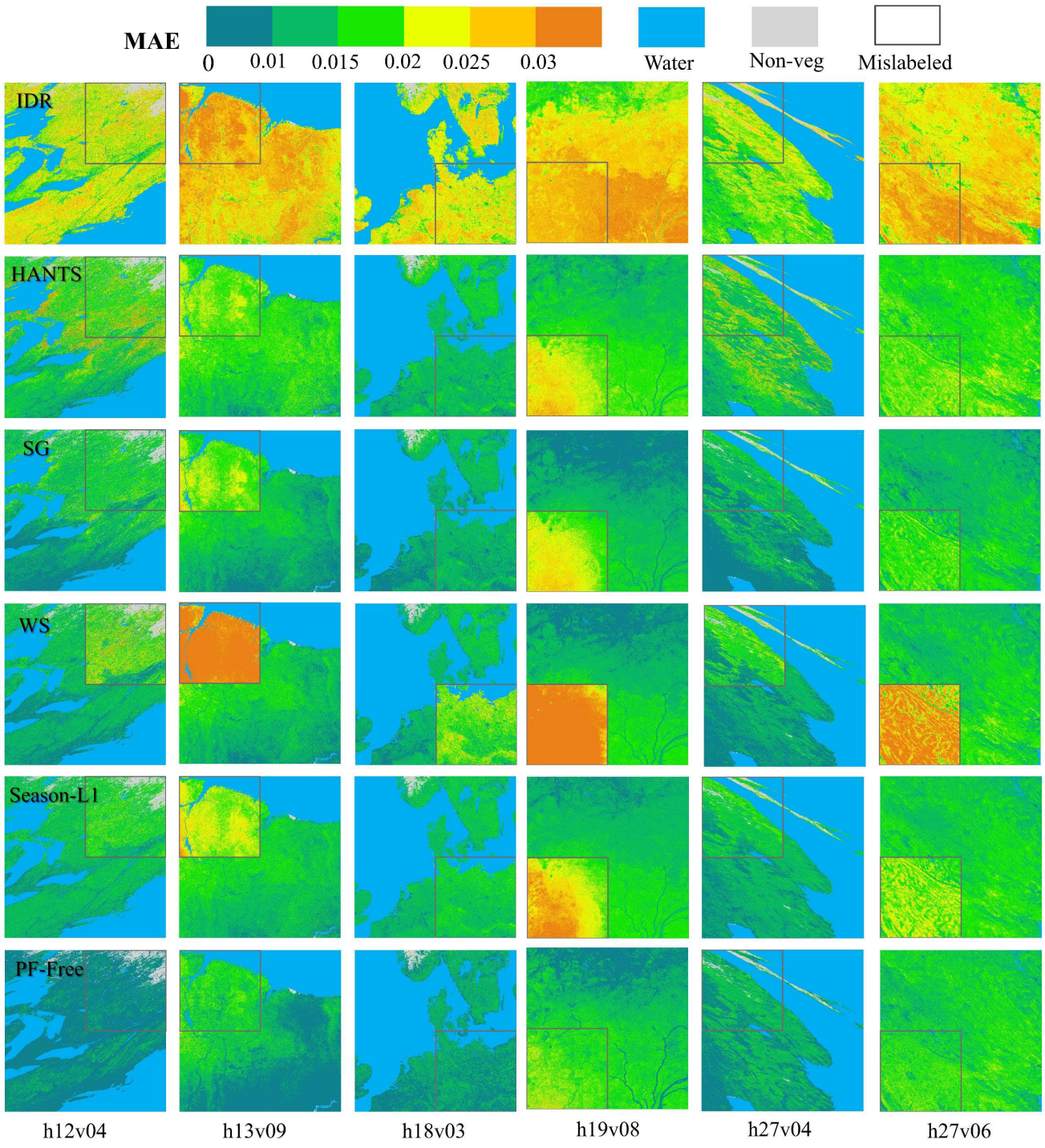


Fig. 4. Spatial distribution of the MAE of the different methods for noise points (the small boxes show the extent of the mislabeled subregions).

smoothing, better restoring the real spatial distribution pattern of the NDVI. Region B is located on the border of Guangxi, Guizhou, and Hunan provinces in China and is mainly covered by forests. Similarly, this region also suffers from severe cloud contamination [see Fig. 8(b)]. Although most of the pixels in the original labeled image are labeled as gaps, there is still a large amount of residual noise in the results of the comparison methods [see the black circles in Fig. 8(b)]. This may be because there are gap points in the adjacent time phases labeled as valid or uncertain points, leading to the failure of reconstruction in this period. In contrast, PF-Free adaptively generates a set of accurate quality flags and thus successfully recovers the gaps.

C. Ablation Experiments

1) Importance of Accurate Quality Flags: Limited by the accuracy of the cloud detection methods, the quality flags

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

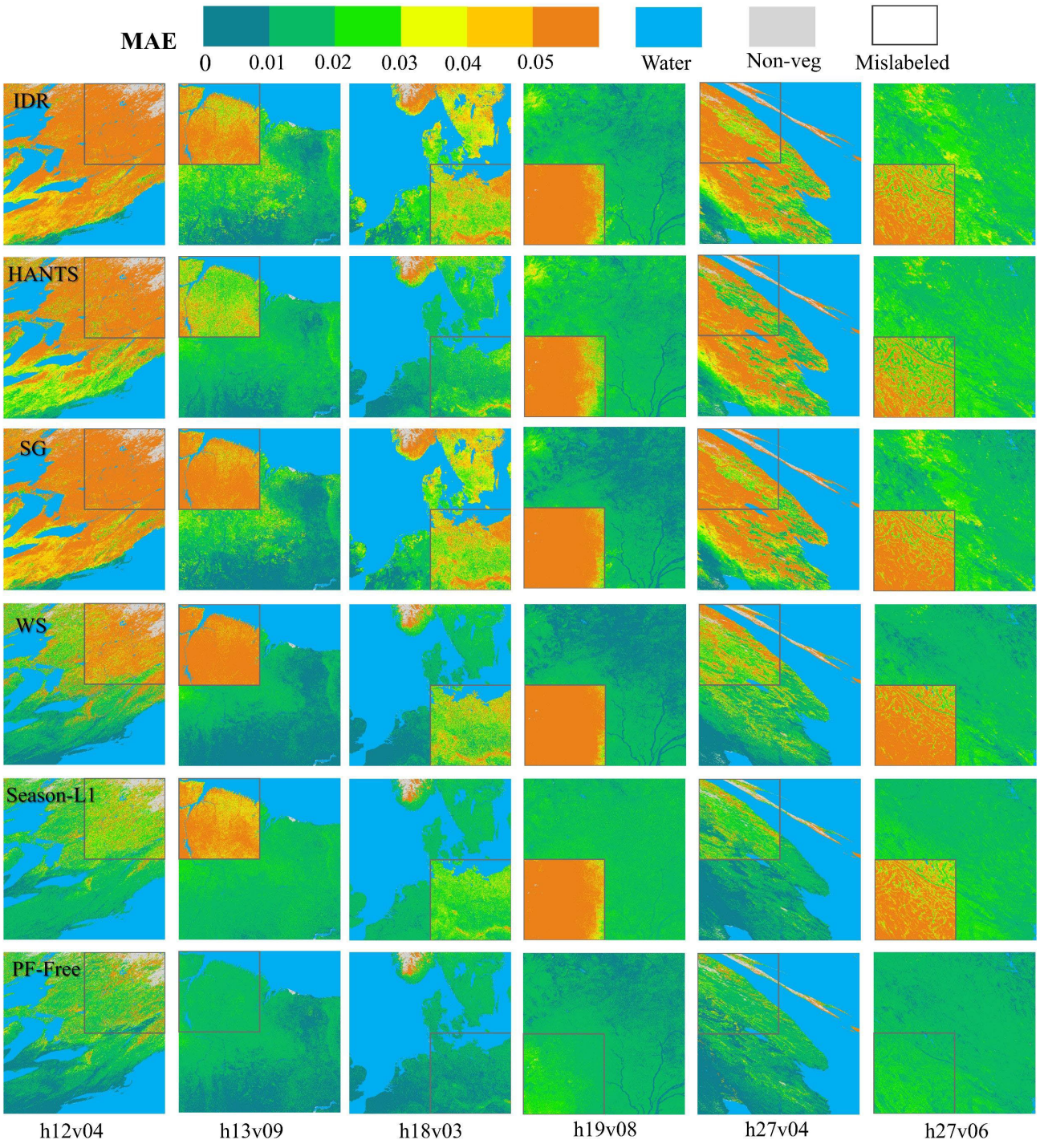


Fig. 5. Spatial distribution of the MAE of the different methods for gap points (the small boxes show the extent of the mislabeled subregions).

provided by VI products are not fully reliable. Therefore, adaptive flag generation is performed before VI reconstruction to eliminate the misleading impacts of inaccurate quality flags. However, the impact of inaccurate quality flags on the reconstruction results has not yet been quantified. To further investigate this relationship, an additional experiment was designed in this study. Specifically, 1000 reference NDVI time-series points were randomly selected in each tile region, simulated as contaminated time series, and then artificially generated mislabeling was applied in the same way as in the simulation experiments, with only the setting of the error ratio varying. In this experiment, we increased the mislabeling ratio from 0% to 50% in steps of 2% to obtain 26 different sets of inaccurate quality flags. Using the PF-Free method, reconstruction was performed using the 26 sets of quality flags and the self-labeled flags. The MAE and CC values for all the points are shown in Fig. 9. When the mislabeling ratio is 0, which represents correct flags, the MAE and CC values

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

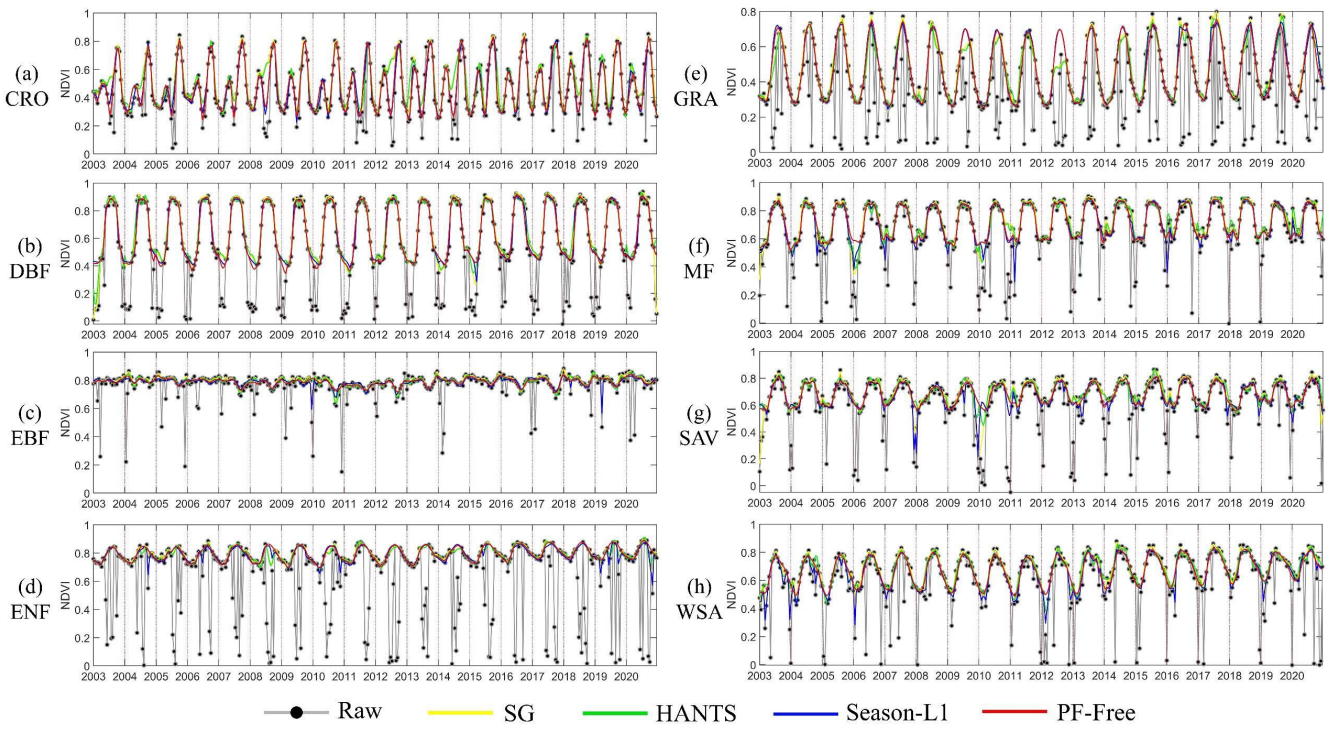


Fig. 6. Comparison of the temporal curves of the different reconstruction methods on eight typical vegetation pixels. (a) CRO. (b) DBF. (c) EBF. (d) ENF. (e) GRA. (f) MF. (g) SAV. (h) WSA.

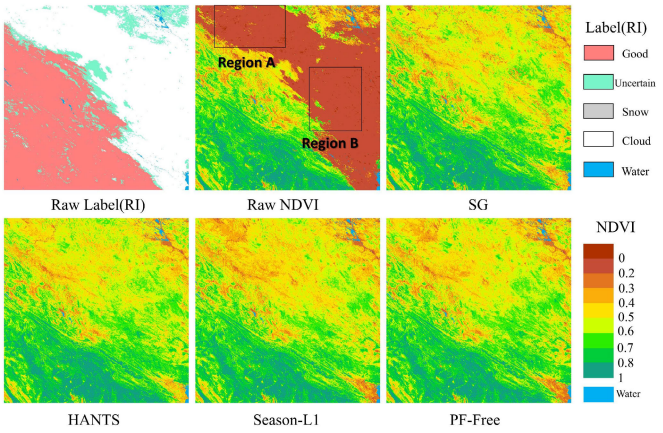


Fig. 7. Comparison of the spatial images of different reconstruction methods in the h27v06 tile region for DOY 1 in 2012.

obtained using both the original correct flags and the adaptive labels are very close to each other, which directly indicates that self-labeling quality flags by the adaptive method is reliable. As the ratio increases, the accuracy using mislabeled flags shows a rapidly decreasing trend. For example, when the mislabeling ratio reaches 50%, the MAE of all the points is 0.0663, indicating that the average error of each point in the reconstructed time series is more than 0.06. Although such a high proportion of mislabeling is unlikely to occur in real quality flags, Figs. 4 and 5 visually reflect the serious impact of mislabeling on the reconstruction results. Even when the mislabeling ratio is around 10%, the reconstructed NDVI clearly deviates from the true values. This further demonstrates the importance of accurate quality flags and the significant advantages of the adaptive method.

2) Superiority of Adaptive Parameter Selection: In Sections IV-A and IV-B, we described how we used the PF-Free method with adaptive parameter selection and achieved a superior result. However, the contribution of adaptive parameter selection was not investigated separately in the previous experiments, so does adaptive parameter selection have an advantage over the use of fixed parameters? In this section, we describe how this question was answered through a simulation experiment. Similarly, 6000 reference and simulated NDVI time-series points were generated as described in Section IV-C1. Reconstruction was then performed using (9) with adaptive and fixed parameters using the raw correct flags and the self-labeled flags. When using fixed parameters, the values of parameters k and λ started at 0.1 and were increased in steps of 0.1–1.5. Finally, the MAE and CC values of the noise and gap points were calculated together, as shown in Fig. 10. Under both the raw correct flags scenario and the self-labeled flags scenario, the adaptive parameter selection approach achieves lower MAE and higher CC values than the use of fixed parameters, indicating that the reconstruction accuracy with adaptive parameter selection is basically better than that of fixed parameters in both cases.

The reason why the adaptive parameter selection approach performs well is that it can automatically select the appropriate regularization parameters according to the characteristics of each time series. This is very beneficial for the task of reconstructing large or complex areas, where the reconstruction parameters are usually strongly heterogeneous. Fig. 11 shows the spatial distribution of parameters k, λ, and vegetation types

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

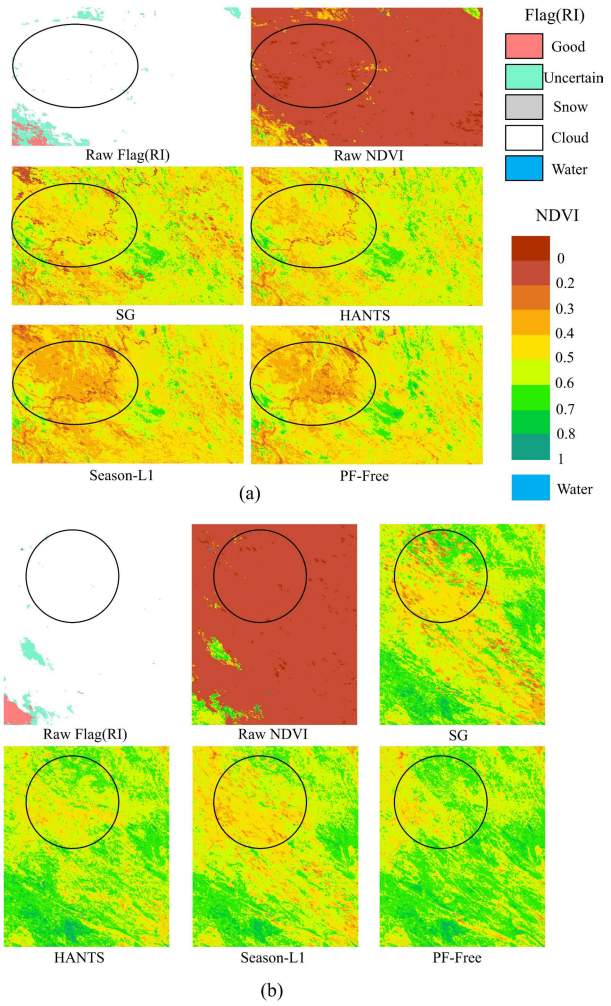


Fig. 8. Comparison of the enlarged spatial images of different reconstruction methods in regions A and B in Fig. 7 for DOY 1 in 2012 (the areas covered by black circles indicate regions where significant differences exist among these methods). (a) Region A. (b) Region B.

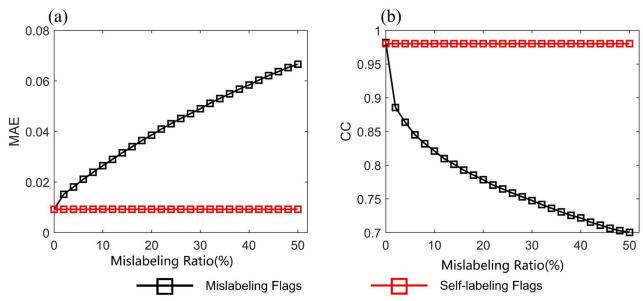


Fig. 9. Variations in MAE and CC with different mislabeling ratios for the PF-Free method. (a) MAE. (b) CC.

in 2020 when reconstructing the real NDVI time series for the h27v06 tile region. It can be seen that the values of parameters k and λ are related to the vegetation type. In forested areas, such as evergreen broadleaf forests, k does not usually exceed 0.1, while λ is mostly greater than 1. This is because vegetation growth in these regions is less subject to human intervention, and their NDVI trends are essentially driven only by natural factors such as climate change. Therefore, the interannual similarity constraint is generally weaker than the temporal smoothness constraint. At the same time, forested areas tend to be cloudy and rainy, and there is more noise and more gaps in the time series, so a larger value of λ is needed to achieve gap filling. However, cropland is more influenced by human activities and its growth is more regular, with k mostly above 0.5 and λ less than 0.5. When fixed parameters are used for reconstruction, although leading to better results overall, it still performs poorly in localized areas. Furthermore, it is difficult to overcome the contradiction between the high computational cost of determining the parameters and their unrepresentative nature.

D. *Applications of the PF-Free Method*

The main innovation of this study is the adaptive generation of quality flags and reconstruction parameters. To emphasize the usefulness of the PF-Free method in practical reconstruction applications, two real-data experiments were conducted in this study: reconstruction of AVHRR GIMMS NDVI3g+ and daily MOD09GA-NDVI datasets. Among them, there are no flag data for GIMMS NDVI3g+, and the flag for MOD09GA may be unreliable due to the high-frequency data. Therefore, it is quite challenging to reconstruct them. The experimental procedures and results are described below.

1) Reconstruction of AVHRR GIMMS NDVI3g+: In this section, we describe how the AVHRR GIMMS NDVI3g+ dataset for the equatorial region of Africa during 1982–2022 was selected and reconstructed using the SG filter, HANTS method, and PF-Free method, where the quality flags were not additionally introduced when using the SG filter and the HANTS method. The temporal curves for three typical pixels are shown in Fig. 12, where the quality flags are those generated by PF-Free. As can be seen in Fig. 12, during the period of high-quality NDVI data, the reconstructed curves of the three methods do not differ much, even without the assistance of quality flags. In contrast, in the presence of noise or gaps, the results of the SG filter and HANTS method show obvious anomalous fluctuations, while the PF-Free method can obtain a smoother reconstruction curve. This can be attributed to the fact that PF-Free uses accurately generated quality flags.

Fig. 13 compares the original image from early December 2014 with the reconstructed images from the three methods. Overall, the reconstruction results of the three methods are comparable but are inconsistent in the southeast region (closed box area). There is a lot of small-scale noise contamination in the original image, and the SG filter and HANTS method find it difficult to smooth this, resulting in a lack of spatial continuity in their reconstruction images. Only PF-Free produces smoother and more complete results. It is, therefore, confirmed that the adaptive flag generation approach is capable of dealing with scenarios without quality flags and ultimately yields more robust results.

2) Reconstruction of Daily MOD09GA-NDVI: In this section, we describe how the MOD09GA-NDVI dataset for a 100 × 100 km area of central Yunnan province, China, for the period 2016–2021 was selected and provide the results obtained with the PF-Free method, SG filter, and HANTS method. Considering the differences in temporal resolution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

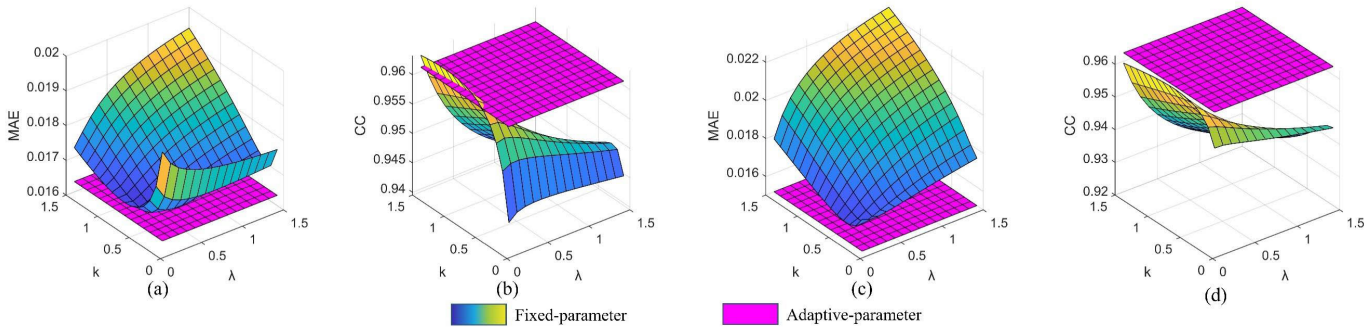


Fig. 10. MAE and CC values for fixed parameters and the adaptive parameter selection approach under the raw correct flags scenario and the self-labeled flags scenario. (a) MAE for raw correct flags. (b) CC for raw correct flags. (c) MAE for self-labeling flags. (d) CC for self-labeling flags.

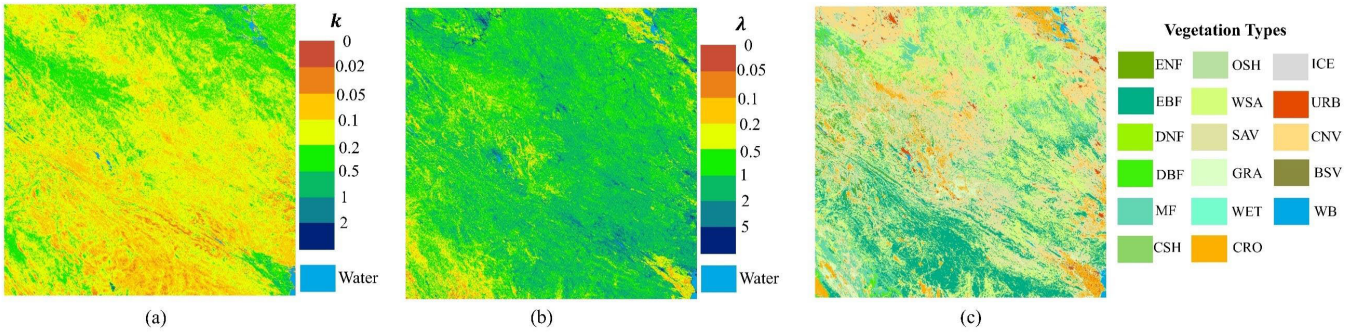


Fig. 11. Spatial distribution of parameters k, λ, and vegetation types in 2020 when reconstructing the real NDVI time series for the h27v06 tile region. (a) k, (b) λ, and (c) vegetation types. ENF, evergreen needleleaf forests; EBF, evergreen broadleaf forests; DNF, deciduous needleleaf forests; DBF, deciduous broadleaf forests; MF, mixed forests; CSH, closed shrublands; OSH, open shrublands; WSA, woody savannas; SAV, savannas; GRA, grasslands; WET, permanent wetlands; CRO, croplands; URB, urban and built-up lands; CNV, cropland/natural vegetation mosaics; ICE, permanent snow and ice; BSV, barren; WB, water bodies.

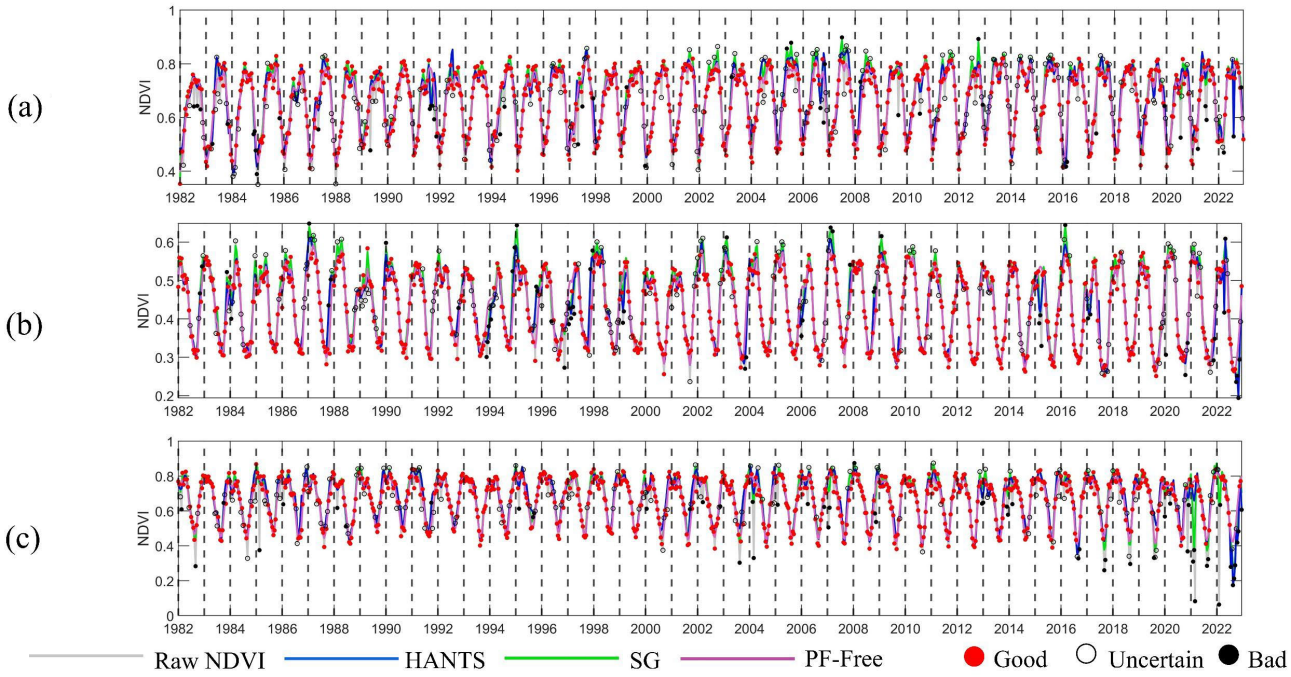


Fig. 12. Comparison of the temporal curves of different reconstruction methods for the AVHRR GIMMS NDVI3g+ dataset during 1982–2022. (a) WSA. (b) GRA. (c) MF.

and length of the MOD09GA-NDVI and MOD13A2-NDVI datasets, the adaptive flag generation approach needed to be adjusted, mainly by the feature point occupancy and the order of curve fits in Table II in supplementary material. Similarly,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

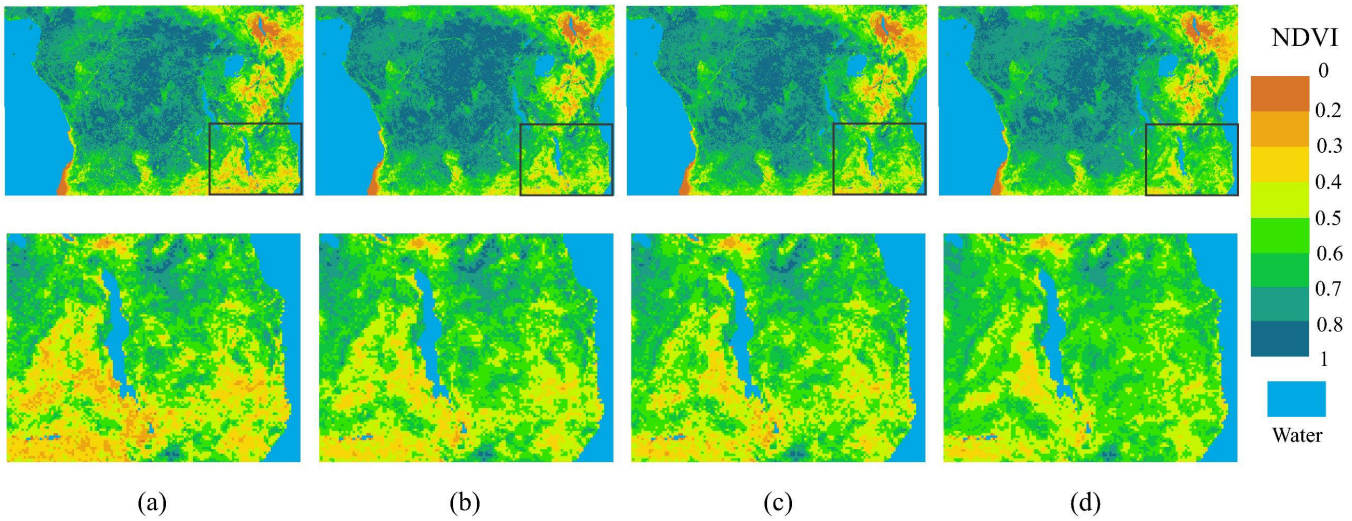


Fig. 13. Comparison of spatial images of the different reconstruction methods for the AVHRR GIMMS NDVI3g+ dataset in West Africa, early December 2014 (the areas covered by closed boxes in the first row indicate regions where significant differences exist among these methods, and the second row of images are the result of zooming in with closed boxes covering the images in the first row). (a) Raw NDVI. (b) SG. (c) HANTS. (d) PF-Free.

the NDVI temporal curves were first plotted for the three typical image elements (see Fig. 14, with quality flags obtained by the adaptive method). The raw MOD09GA-NDVI time series is quite noisy and has missing data, with which the SG filter is unable to achieve reconstruction, while the HANTS method results in a general overestimation of the NDVI values. On the other hand, the proposed PF-Free method is still able to restore the true NDVI growth trajectory in this extreme scenario. This can be explained by the reasonable choice of reconstruction parameters, which allows the curve to strike a balance between smoothing and detail preservation.

Fig. 15 compares the images before and after MOD09GA-NDVI reconstruction for DOY 70 in 2017 and DOY 150 in 2021. Due to cloud contamination, there are numerous low values in the original images, which are basically unusable. The SG filter struggles to remove the low values completely, and the HANTS method shows dense high-value noise. The PF-Free method is the only method able to restore the normal distribution pattern of the NDVI, illustrating its excellent performance in the daily VI reconstruction task.

V. Discussion

A. Advantages of PF-Free

High-quality VI data are of great importance for vegetation remote sensing research. Therefore, a large number of reconstruction methods for the VI time series have been developed based on different perspectives over the past decades. However, these methods all face two major difficulties, namely, the dependence on quality flags and the setting of reconstruction parameters. Absent or inaccurate quality flags and inappropriate reconstruction parameters greatly affect the accuracy of the reconstruction methods, limiting their ability to be applied to large and complex regions. In view of this, an adaptive reconstruction method without requiring the input of quality flags or parameters (PF-Free) has been proposed in this article, providing a more robust and convenient solution that is more practical for large-scale applications. The advantages of the PF-Free method are reflected in two main aspects.

First of all, PF-Free can generate plausible quality flags through the adaptive flag generation approach, thus avoiding the impact of missing or inaccurate quality flags on the reconstruction. In practice, most of the existing methods rely on quality flags to aid the reconstruction. In scenarios where flags are not provided (e.g., AVHRR GIMMS NDVI3g+), especially when there are large gaps in the time series, the precision will be drastically diminished. As for PF-Free, considering that different types of vegetation generally follow the same cycle pattern, the self-labeling step allows for the rapid generation of corresponding quality flags based on the cyclical pattern of vegetation growth, which in turn allows for effective reconstruction in unlabeled situations (Figs. 12 and 13). Furthermore, even when raw quality flags are provided, there can be uncertainty in their quality [31]. This is a normal phenomenon in the field of vegetation remote sensing. Given the limited accuracy of algorithms such as cloud detection algorithms, it is difficult to precisely mark the quality status of each pixel point. As can be seen in Fig. 6, the reconstruction results of the comparison methods are all abnormally distorted after using the wrong flags. In such poor conditions for reconstruction, the self-labeled flags can be more accurate than the original flags, and the results are naturally more reflective of the actual growth status of the vegetation.

Another advantage is that PF-Free can automatically determine the optimal parameters on a pixel-by-pixel basis, which is extremely flexible for large-scale reconstruction tasks. Reconstruction parameter settings are generally related to noise and the missing content of the data, pixel vegetation type, and other factors. For the commonly used classical reconstruction methods, such as the HANTS method, the SG filter, and the WS, studies have been conducted to determine the reconstruction

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

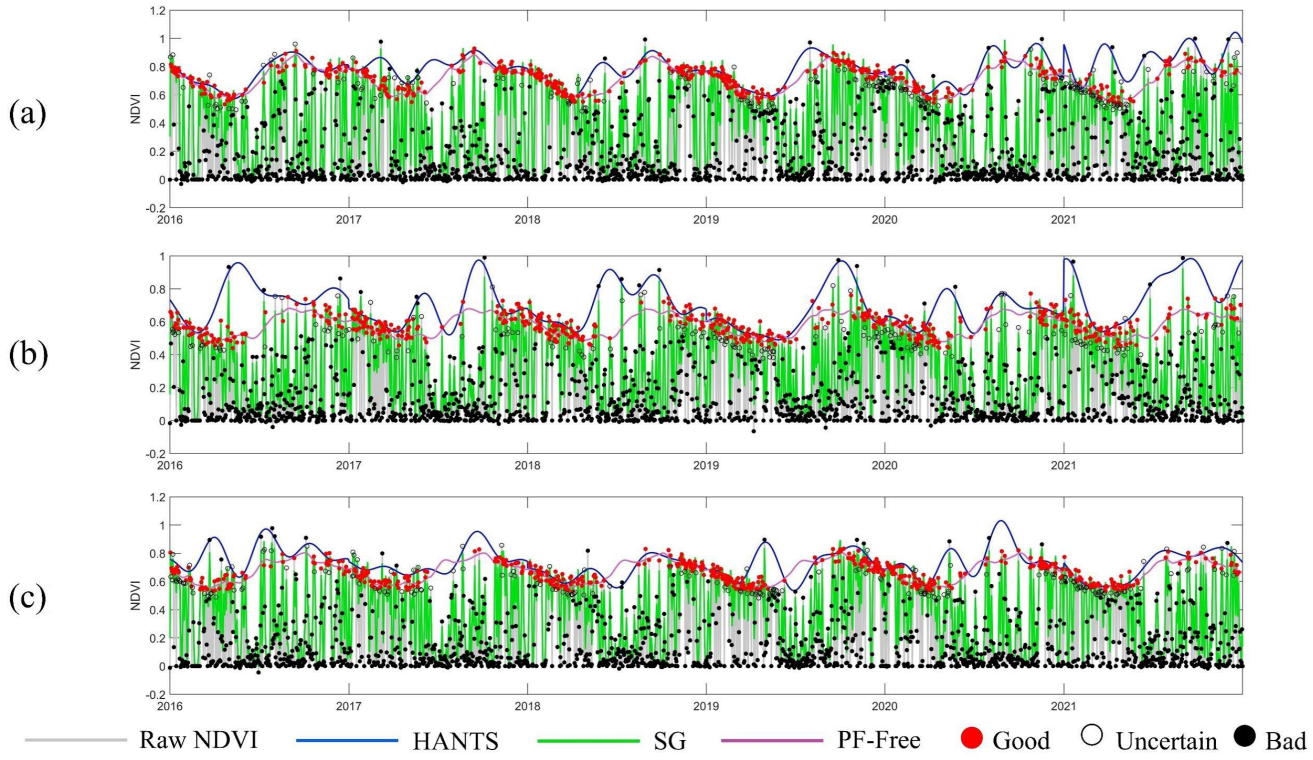


Fig. 14. Comparison of the temporal curves of different reconstruction methods for the MOD09GA-NDVI dataset during 2016–2021. (a) EBF. (b) SAV. (c) MF.

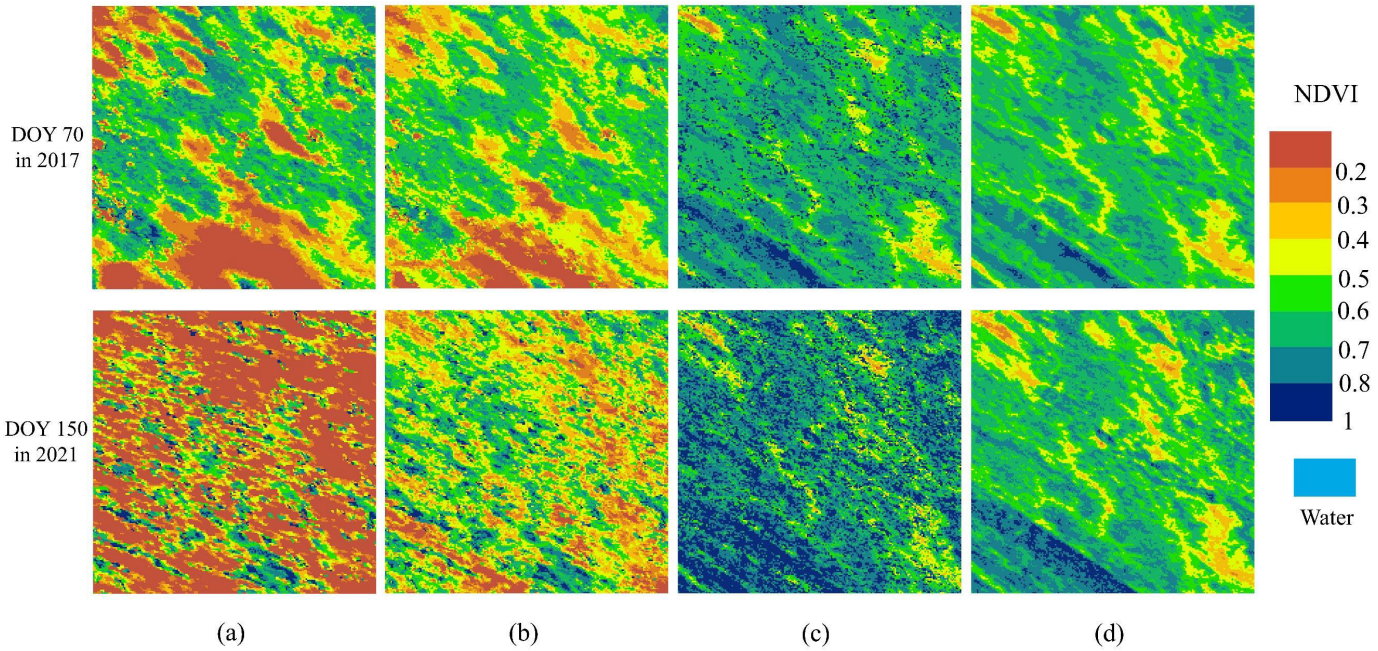


Fig. 15. Comparison of the spatial images of different reconstruction methods with the MOD09GA-NDVI dataset for the 100 × 100 km area of central Yunnan province for DOY 70 in 2017 and DOY 150 in 2021. (a) Raw NDVI. (b) SG. (c) HANTS. (d) PF-Free.

parameters of the above methods through global-scale experiments [27], [40]. These parameters remain uniform and fixed and, in general, better reconstruction results can be obtained using such fixed parameters. However, owing to the strong spatial heterogeneity of VIs, the more different the time series, the greater the variation in the reconstruction parameters. Applying fixed parameters to a large reconstruction task can lead to good results in some areas and poor results in others. The PF-Free method is based on the improved GCV technique for adaptive parameter selection, and for each time

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

series, unique parameters can be determined automatically. The superiority of the adaptive parameters over the fixed parameters in terms of reconstruction performance is shown in Fig. 10.

In summary, PF-Free will have important applications in large-scale geo-analytical studies. On large regions such as the globe, since vegetation types and meteorological conditions are variable, the quality of land surface data such as VI data is unstable. This highly adaptive method can quickly perform the task of reconstructing the above areas, thus providing reliable and effective data support for the analysis of relevant conclusions. For example, PF-Free can enhance the quality of long time-series data from sensors like AVHRR and MODIS. This improvement raises the accuracy of downstream applications such as vegetation productivity estimation and climate extraction [60], [61]. In addition, PF-Free is suitable for reconstructing other data like leaf area index (LAI) and surface reflectance, as well as the fine-resolution data from different satellite sensors, given that they follow the same regularization prior [62]. Figs. 1 and 2 in the supplementary material show the results of reconstructing Landsat NDVI data and MODIS daily surface reflectance using the PF-Free method, from which promising applicability can be observed. Moreover, considering that the PF-Free method based on two-paradigm constraints has well-defined code portability, PF-Free can be deployed on Google Earth Engine (GEE) in the future, which will combine GEE's powerful data management capabilities [63] with this robust and convenient VI reconstruction method. Users will only need to retrieve satellite images, and then directly use PF-Free to reconstruct without any additional settings.

B. Limitations

Despite the satisfactory performance of PF-Free in reconstructing VIs at large scales due to which it can obtain parameters and flags adaptively, this method still has some limitations in some situations. First of all, the adaptive flag generation method is unable to identify the specific type or level of gap points, such as with both cloud cover and snow cover all considered to be gaps. This is mainly because this step only determines the quality flags of the data points by their spatial relationship to the fit confidence interval based on mathematical statistics. Points outside the confidence interval are more likely to be gap points than those inside the confidence interval, and the specific type cannot be obtained because no other information is introduced. Although the delineation of cloud-covered and snow-covered points is crucial for some applications like vegetation phenology monitoring [25], this will ultimately not affect the reconstruction results even if the two flags are not distinguished. In the future, the introduction of other auxiliary variables, such as meteorological data, will be considered to fully distinguish the specific types of missing points [64]. In addition, it is difficult for PF-Free to accurately label a time series with obvious land-cover changes. PF-Free usually assumes similar values for the same time phase over several years, however, human activities or natural disasters-induced land-cover changes, including deforestation, forestation, or vegetation type conversion, would bring uncertainties into the results [65], [66]. As dramatic land-cover change usually induces abnormally low NDVI values when it occurs, it would tend to label these low-value data for each time phase as gaps. There are currently two possible solutions: one is to label each year several times using a sliding window and take the flag with the highest number of occurrences from the multiple sets of results as the final. It is also possible to identify the temporal nodes of land cover change and use them as boundaries for segmented self-labeling. Fig. 3 in the supplementary material illustrates that the PF-Free method of both solutions can deal effectively with the challenge of vegetation type conversion. Furthermore, similar to many other traditional methods, PF-Free also suffers from noise and anomalies of episodic extremity, such as volcanic eruptions or military interferences [67]. If the anomalies have a persistent effect on the vegetation, or similar anomalies occurred many years ago, then the proposed PF-Free can take advantage of flexible parameter settings to adjust the strength of the constraints on temporal smoothness and interannual similarity for reconstruction (see Fig. 4 in the supplementary material). However, it would certainly be challenging if such sudden anomalies resulted in sharp and permanent NDVI changes in vegetation. Finally, the computational cost of PF-Free deserves to be re-optimized, with the efficiency (14.18 s for average result under the MATLAB environment on a desktop computer with an i7-13700 CPU (2.10 GHz) and 32 GB of RAM) slightly slower than that of the Season-L1 method (11.77 s) [54]. In the case of monitoring croplands all over the world, take the input of three-year MOD09CMG-NDVI daily data as an example. It takes about 6 h to reconstruct all the croplands, which we think can generally meet the practical application requirements of the near real-time task. The time cost is mainly concentrated in the adaptive parameter selection, and we found the time cost can be significantly decreased through segmental reconstruction (see Fig. 5 in the supplementary material) with stable accuracy. As a result, it is directed to divide the long time series into different parts for reconstruction if you want to improve the time efficiency of PF-Free. Further efforts are also directed to enhance the process efficiency of PF-Free by parallelized computation and cloud platforms.

VI. Conclusion

In the field of VI reconstruction, quality flag dependence and reconstruction parameter setting have been two major difficulties common to most methods. In this article, we have proposed a parameter-free and flag-free adaptive reconstruction method (PF-Free) that requires only the input of the raw VI time series and fully exploits the cyclical pattern of vegetation to generate quality flags. Meanwhile, the improved GCV technique is innovatively introduced into the reconstruction parameter selection. The experiments confirmed that the PF-Free method excels in both quantitative metrics and qualitative visualization, which can be attributed to its two main strengths. On the one hand, with the help of the adaptive flag generation method, this method can directly avoid the impacts of missing or wrong quality flags on the VI reconstruction.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

On the other hand, the convenient and flexible strategy of adaptive parameter selection allows it to select the reconstruction parameters pixel by pixel, which, in turn, provides greater robustness. Further experiments also demonstrated the effectiveness of PF-Free in scenarios without quality flags or with severely contaminated data. We believe that this practical approach will be useful in future large-scale reconstruction tasks, contributing to the generation of high-quality VI time series and subsequent analytical applications. The main code of the PF-Free method is available on https://github.com/ranyuxi/PF-Free for free.

ACKNOWLEDGMENT

The authors would like to thank NASA for free access to MODIS products and Pinzon et al. for providing GIMMS NDVI3g+ products. They also appreciate Dr. Dong Chu for providing the codes and detailed guidance on the Season-L1 method.

REFERENCES

[1] Y. Zeng et al., "Optical vegetation indices for monitoring terrestrial ecosystems globally," *Nature Rev. Earth Environ.*, vol. 3, no. 7, pp. 477–493, May 2022.

[2] R. B. Myneni and D. L. Williams, "On the relationship between FAPAR and NDVI," *Remote Sens. Environ.*, vol. 49, no. 3, pp. 200–211, Sep. 1994.

[3] X. Guan, H. Shen, X. Li, W. Gan, and L. Zhang, "A long-term and comprehensive assessment of the urbanization-induced impacts on vegetation net primary productivity," *Sci. Total Environ.*, vol. 669, pp. 342–352, Jun. 2019.

[4] S. Peng et al., "Asymmetric effects of daytime and night-time warming on northern Hemisphere vegetation," *Nature*, vol. 501, no. 7465, pp. 88–92, Sep. 2013.

[5] X. Yang, J. Chen, Q. Guan, H. Gao, and W. Xia, "Enhanced spatial–temporal Savitzky–Golay method for reconstructing high-quality NDVI time series: Reduced sensitivity to quality flags and improved computational efficiency," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4412317.

[6] B. N. Holben, "Characteristics of maximum-value composite images from temporal AVHRR data," *Int. J. Remote Sens.*, vol. 7, no. 11, pp. 1417–1434, Nov. 1986.

[7] L. Geng, M. Ma, X. Wang, W. Yu, S. Jia, and H. Wang, "Comparison of eight techniques for reconstructing multi-satellite sensor time-series NDVI data sets in the Heihe River Basin, China," *Remote Sens.*, vol. 6, no. 3, pp. 2024–2049, Mar. 2014.

[8] Y. Sun, D. Peng, X. Guan, D. Chu, Y. Ma, and H. Shen, "Impacts of the data quality of remote sensing vegetation index on gross primary productivity estimation," *GISci. Remote Sens.*, vol. 60, no. 1, Dec. 2023, Art. no. 2275421.

[9] Y. Shao, R. S. Lunetta, B. Wheeler, J. S. Iiames, and J. B. Campbell, "An evaluation of time-series smoothing algorithms for land-cover classifications using MODIS-NDVI multi-temporal data," *Remote Sens. Environ.*, vol. 174, pp. 258–265, Mar. 2016.

[10] F. Xie and H. Fan, "Deriving drought indices from MODIS vegetation indices (NDVI/EVI) and Land Surface Temperature (LST): Is data reconstruction necessary?" *Int. J. Appl. Earth Observ. Geoinf.*, vol. 101, Sep. 2021, Art. no. 102352.

[11] Y. Chen, R. Cao, J. Chen, L. Liu, and B. Matsushita, "A practical approach to reconstruct high-quality Landsat NDVI time-series data by gap filling and the Savitzky–Golay filter," *ISPRS J. Photogramm. Remote Sens.*, vol. 180, pp. 174–190, Oct. 2021.

[12] R. Cao et al., "A simple method to improve the quality of NDVI time-series data by integrating spatiotemporal information with the Savitzky–Golay filter," *Remote Sens. Environ.*, vol. 217, pp. 244–257, Nov. 2018.

[13] D. Chu et al., "Long time-series NDVI reconstruction in cloud-prone regions via spatio-temporal tensor completion," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112632.

[14] S. Li, L. Xu, Y. Jing, H. Yin, X. Li, and X. Guan, "High-quality vegetation index product generation: A review of NDVI time series reconstruction techniques," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102640.

[15] H. Shen et al., "Missing information reconstruction of remote sensing data: A technical review," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 61–85, Sep. 2015.

[16] P. S. A. Beck, C. Atzberger, K. A. Høgda, B. Johansen, and A. K. Skidmore, "Improved monitoring of vegetation dynamics at very high latitudes: A new method using MODIS NDVI," *Remote Sens. Environ.*, vol. 100, no. 3, pp. 321–334, Feb. 2006.

[17] P. Jonsson and L. Eklundh, "Seasonality extraction by function fitting to time-series of satellite sensor data," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 8, pp. 1824–1832, Aug. 2002.

[18] J. Chen, P. Jonsson, M. Tamura, Z. H. Gu, B. Matsushita, and L. Eklundh, "A simple method for reconstructing a high-quality NDVI time-series data set based on the Savitzky–Golay filter," *Remote Sens. Environ.*, vol. 91, nos. 3–4, pp. 332–344, Jun. 2004.

[19] Y. Julien and J. A. Sobrino, "Comparison of cloud-reconstruction methods for time series of composite NDVI data," *Remote Sens. Environ.*, vol. 114, no. 3, pp. 618–625, Mar. 2010.

[20] M. Menenti, S. Azzali, W. Verhoef, and R. van Swol, "Mapping agroecological zones and time lag in vegetation growth by means of Fourier analysis of time series of NDVI images," *Adv. Space Res.*, vol. 13, no. 5, pp. 233–237, May 1993.

[21] G. Yang, H. Shen, L. Zhang, Z. He, and X. Li, "A moving weighted harmonic analysis method for reconstructing high-quality SPOT VEGETATION NDVI time-series data," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 11, pp. 6008–6021, Nov. 2015.

[22] C. Atzberger and P. H. C. Eilers, "A time series for monitoring vegetation activity and phenology at 10-daily time steps covering large parts of South America," *Int. J. Digit. Earth*, vol. 4, no. 5, pp. 365–386, Sep. 2011.

[23] X. Liu, H. Shen, Q. Yuan, X. Lu, and S. Li, "One-step high-quality NDVI time-series reconstruction by joint modeling of gradual vegetation change and negatively biased atmospheric contamination," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4407017.

[24] N. Viovy, O. Arino, and A. S. Belward, "The best index slope extraction (BISE): A method for reducing noise in NDVI time-series," *Int. J. Remote Sens.*, vol. 13, no. 8, pp. 1585–1590, May 1992.

[25] M. Shen et al., "Challenges in remote sensing of vegetation phenology," *Innov. Geosci.*, vol. 2, no. 2, Jan. 2024, Art. no. 100070.

[26] C. Sun, Z. Xue, L. Zhang, and H. Su, "Local peak Savitzky–Golay for spatio-temporal reconstruction of Landsat NDVI time series: A case study over the Qinghai–Tibet Plateau," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13439–13455, 2024.

[27] J. Zhou, L. Jia, M. Menenti, and B. Gorte, "On the performance of remote sensing time series reconstruction methods—A spatial comparison," *Remote Sens. Environ.*, vol. 187, pp. 367–384, Dec. 2016.

[28] T. Hilker, A. I. Lyapustin, C. J. Tucker, P. J. Sellers, F. G. Hall, and Y. Wang, "Remote sensing of tropical ecosystems: Atmospheric correction and cloud masking matter," *Remote Sens. Environ.*, vol. 127, pp. 370–384, Dec. 2012.

[29] R. Liu and Y. Liu, "Generation of new cloud masks from MODIS land surface reflectance products," *Remote Sens. Environ.*, vol. 133, pp. 21–37, Jun. 2013.

[30] K. Grogan and R. Fensholt, "Exploring patterns and effects of aerosol quantity flag anomalies in MODIS surface reflectance products in the tropics," *Remote Sens.*, vol. 5, no. 7, pp. 3495–3515, Jul. 2013.

[31] E. Masuoka et al., "MODIS land data products: Generation, quality assurance and validation," in *Land Remote Sensing and Global Environmental Change: NASA's Earth Observing System and the Science of ASTER and MODIS*, B. Ramachandran, C. O. Justice, and M. J. Abrams, Eds., New York, NY, USA: Springer, 2011, pp. 509–531.

[32] D. P. Roy, J. S. Borak, S. Devadiga, R. E. Wolfe, M. Zheng, and J. Descloitres, "The MODIS land product quality assessment approach," *Remote Sens. Environ.*, vol. 83, nos. 1–2, pp. 62–76, Nov. 2002.

[33] M. Cai, Y. Zhang, X. Guan, and J. Qiu, "An adaptive spatiotemporal tensor reconstruction method for GIMMS-3g+ NDVI," *Remote Sens. Environ.*, vol. 316, Jan. 2025, Art. no. 114511.

[34] H. Shen, L. Peng, L. Yue, Q. Yuan, and L. Zhang, "Adaptive norm selection for regularized image restoration and super-resolution," *IEEE Trans. Cybern.*, vol. 46, no. 6, pp. 1388–1399, Jun. 2016.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.

[35] H. Shen and L. Zhang, "A MAP-based algorithm for destriping and inpainting of remotely sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 5, pp. 1492–1502, May 2009.

[36] Z. Cai, P. Jönsson, H. Jin, and L. Eklundh, "Performance of smoothing methods for reconstructing NDVI time-series and estimating vegetation phenology from MODIS data," *Remote Sens.*, vol. 9, no. 12, p. 1271, Dec. 2017.

[37] D. Li, D. Chu, X. Guan, W. He, and H. Shen, "Adaptive regularized low-rank tensor decomposition for hyperspectral image denoising and destriping," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5513917.

[38] L. Zhang, H. Shen, W. Gong, and H. Zhang, "Adjustable model-based fusion method for multispectral and panchromatic images," *IEEE Trans. Syst. Man, Cybern. B, Cybern.*, vol. 42, no. 6, pp. 1693–1704, Dec. 2012.

[39] J. Zhou, L. Jia, M. Menenti, and X. Liu, "Optimal estimate of global biome—Specific parameter settings to reconstruct NDVI time series with the harmonic ANalysis of time series (HANTS) method," *Remote Sens.*, vol. 13, no. 21, p. 4251, Oct. 2021.

[40] R. Liu, R. Shang, Y. Liu, and X. Lu, "Global evaluation of gap-filling approaches for seasonal NDVI with considering vegetation growth trajectory, protection of key point, noise resistance and curve stability," *Remote Sens. Environ.*, vol. 189, pp. 164–179, Feb. 2017.

[41] W. Zhu, B. He, Z. Xie, C. Zhao, H. Zhuang, and P. Li, "Reconstruction of vegetation index time series based on self-weighting function fitting from curve features," *Remote Sens.*, vol. 14, no. 9, p. 2247, May 2022.

[42] Y. Zhang, R. Li, and C.-L. Tsai, "Regularization parameter selections via generalized information criterion," *J. Amer. Stat. Assoc.*, vol. 105, no. 489, pp. 312–323, Mar. 2010.

[43] L. Zhuang, C.-H. Lin, M. A. T. Figueiredo, and J. M. Bioucas-Dias, "Regularization parameter selection in minimum volume hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9858–9877, Dec. 2019.

[44] D. Kong, Y. Zhang, X. Gu, and D. Wang, "A robust method for reconstructing global MODIS EVI time series on the Google Earth engine," *ISPRS J. Photogramm. Remote Sens.*, vol. 155, pp. 13–24, Sep. 2019.

[45] X. Li, W. Fu, H. Shen, C. Huang, and L. Zhang, "Monitoring snow cover variability (2000–2014) in the Hengduan mountains based on cloud-removed MODIS products with an adaptive spatio-temporal weighted method," *J. Hydrol.*, vol. 551, pp. 314–327, Aug. 2017.

[46] H. Chen et al., "Assessment and inter-comparison of multi-source high spatial resolution evapotranspiration products over Lancang–Mekong River Basin, Southeast Asia," *Remote Sens.*, vol. 14, no. 3, p. 479, Jan. 2022.

[47] F. de Souza Mendes, D. Baron, G. Gerold, V. Liesenberg, and S. Erasmi, "Optical and SAR remote sensing synergism for mapping vegetation types in the endangered Cerrado/Amazon ecotone of Nova Mutum—Mato Grosso," *Remote Sens.*, vol. 11, no. 10, p. 1161, May 2019.

[48] P. Li, Z. Feng, and C. Xiao, "Acquisition probability differences in cloud coverage of the available Landsat observations over Mainland Southeast Asia from 1986 to 2015," *Int. J. Digit. Earth*, vol. 11, no. 5, pp. 437–450, May 2018.

[49] K. Didan and A. Barreto-Muñoz, "MODIS collection 6.1 (C61) vegetationindex product userguide," Univ. Arizona, Vegetation Index Phenology Lab, Tucson, AZ, USA, 2019. [Online]. Available: https://lpdaac.usgs.gov/documents/621/MOD13_User_Guide_V61.pdf

[50] C. Xiong et al., "Improved global 250 m 8-day NDVI and EVI products from 2000–2021 using the LSTM model," *Sci. Data*, vol. 10, no. 1, p. 800, Nov. 2023.

[51] G. Yang, H. Shen, W. Sun, J. Li, N. Diao, and Z. He, "On the generation of gapless and seamless daily surface reflectance data," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4289–4306, Aug. 2018.

[52] J. E. Pinzon et al., "Vegetation collection global vegetation greenness (NDVI) from AVHRR GIMMS-3G+, 1981–2022," ORNL Distrib. Act. Arch. Center, Oak Ridge, TN, USA, 2023, doi: 10.3334/ORNLDAAC/2187. [Online]. Available: https://daac.ornl.gov/cgi-bin/dsviewer.pl?ds_id=2187

[53] Y. Zhang and A. Ye, "Spatial and temporal variations in vegetation coverage observed using AVHRR GIMMS and Terra MODIS data in the Mainland of China," *Int. J. Remote Sens.*, vol. 41, no. 11, pp. 4238–4268, Jun. 2020.

[54] D. Chu, H. Shen, X. Guan, and X. Li, "An L1-regularized variational approach for NDVI time-series reconstruction considering inter-annual seasonal similarity," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 114, Nov. 2022, Art. no. 103021.

[55] P. Craven and G. Wahba, "Smoothing noisy data with spline functions: Estimating the correct degree of smoothing by the method of generalized cross-validation," *Numerische Math.*, vol. 31, no. 4, pp. 377–403, Dec. 1978.

[56] N. Nguyen, P. Milanfar, and G. Golub, "A computationally efficient superresolution image reconstruction algorithm," *IEEE Trans. Image Process.*, vol. 10, no. 4, pp. 573–583, Apr. 2001.

[57] D. Garcia, "Robust smoothing of gridded data in one and higher dimensions with missing values," *Comput. Statist. Data Anal.*, vol. 54, no. 4, pp. 1167–1178, Apr. 2010.

[58] J. B. B. Beckmann, A. Sparks, J. A. Ward-Williams, M. D. Mantle, A. J. Sederman, and L. F. Gladden, "Generalized cross-validation as a method of hyperparameter search for MTGV regularization," 2023, *arXiv:2311.11442*.

[59] J. Zhou, L. Jia, and M. Menenti, "Reconstruction of global MODIS NDVI time series: Performance of harmonic ANalysis of time series (HANTS)," *Remote Sens. Environ.*, vol. 163, pp. 217–228, Jun. 2015.

[60] Y. Guo et al., "Spatial and temporal changes in vegetation in the Ruoergai Region, China," *Forests*, vol. 12, no. 1, p. 76, Jan. 2021.

[61] D. Zhao et al., "Temporal resolution of vegetation indices and solar-induced chlorophyll fluorescence data affects the accuracy of vegetation phenology estimation: A study using in-situ measurements," *Ecolog. Indicators*, vol. 136, Mar. 2022, Art. no. 108673.

[62] J. Wang et al., "Improving the quality of MODIS LAI products by exploiting spatiotemporal correlation information," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4402319.

[63] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau, and R. Moore, "Google Earth Engine: Planetary-scale geospatial analysis for everyone," *Remote Sens. Environ.*, vol. 202, pp. 18–27, Dec. 2017.

[64] J. Pu et al., "Sensor-independent LAI/FPAR CDR: Reconstructing a global sensor-independent climate data record of MODIS and VIIRS LAI/FPAR from 2000 to 2022," *Earth Syst. Sci. Data*, vol. 16, no. 1, pp. 15–34, Jan. 2024.

[65] H. Liu, P. Gong, J. Wang, N. Clinton, Y. Bai, and S. Liang, "Annual dynamics of global land cover and its long-term changes from 1982 to 2015," *Earth Syst. Sci. Data*, vol. 12, no. 2, pp. 1217–1243, Jun. 2020.

[66] F. Yang, F. He, S. Li, M. Li, and P. Wu, "A new estimation of carbon emissions from land use and land cover change in China over the past 300 years," *Sci. Total Environ.*, vol. 863, Mar. 2023, Art. no. 160963.

[67] K. Yan et al., "A bibliometric visualization review of the MODIS LAI/FPAR products from 1995 to 2020," *J. Remote Sens.*, vol. 2021, pp. 1–20, Mar. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:09:38 UTC from IEEE Xplore. Restrictions apply.